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LEAP: DATA ANAYLSIS AND STATISTICS FINAL ASSIGNMENT UNDERSTANDING DIABETES PROGRESSION

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INTODUCTION

This report aims to present and explain the steps of the procedures carried out in the final project of the course “LEAP: Data Analysis and Statistics”. The project involves the analysis of a real-world health dataset comprising 442 individuals. It includes data understanding and exploration, the examination of relationships between variables, the building and interpretation of regression models, and finally, the evaluation of the analysis and the drawing of conclusions. The goal is to identify which patient characteristics are associated with the progression of diabetes, what the regression models tell us, and what conclusions can reasonably be drawn from the data.

DATA

The dataset contains information such as age, sex, body mass index, blood pressure, and various serum measurements- which serve as the main predictor variables- for 442 patients with diabetes, as well as a quantitative measure of disease progression recorded one year after the initial measurement, which serves as the outcome. The dataset consists of 442 rows (each row representing observations for a single patient) across 11 columns (variables under consideration).

Age, BMI, blood pressure, serum measurements and progression are continuous numerical measurements, which is also confirmed by the histograms showing the distribution of their values. Sex is a categorical variable, despite the fact that it is recorded numerically, to suggest that it should be treated as a two-level variable. The data is characterized by very satisfactory quality and completeness, as neither missing values (sum(is.na (diabetes)) returns 0) nor duplicate records (sum(duplicated(diabetes)) returns 0) are observed. At the same time, age, BMI and blood pressure fall within realistic ranges, which indicates that no outliers will disrupt the analysis.

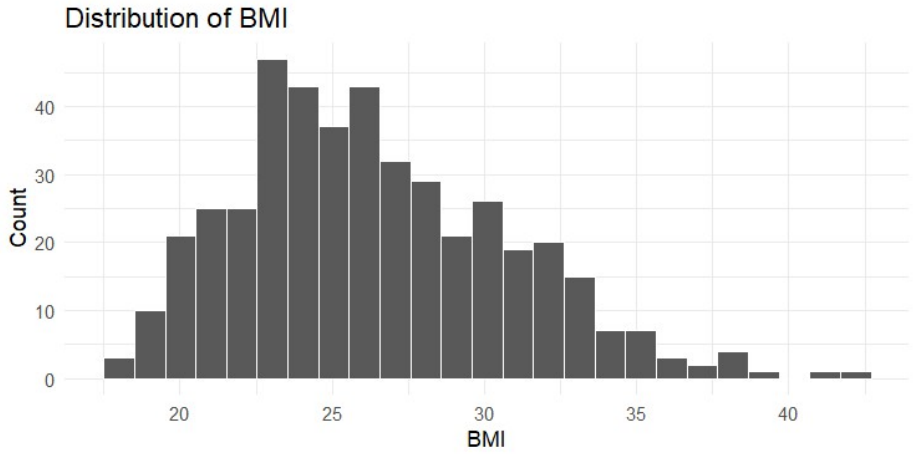
It is important to mention that sensitive data such as health information needs to be protected from any kind of exposure, while our analysis and interpretation must be responsible and very cautious, as people's health depends on them.

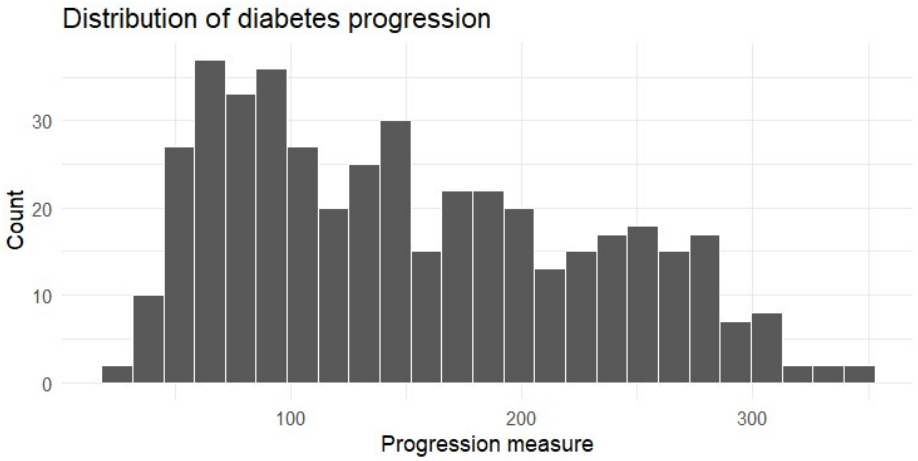
After an initial assessment of the data- both in terms of its characteristics and its completeness- we calculate certain metrics that we need to know before proceeding with the use of the data. The metrics fall within normal ranges, as expected, since no issues were observed in the dataset. To visualize the results, a histogram is plotted to illustrate the distribution of BMI values, and a graph is also plotted to describe the distribution of the outcome, namely progression. Both confirm the absence of problematic data points in the dataset, as evidenced so far.

While predictor values are physiologically plausible and free of distorting anomalies, the rightward skew in progression highlights the need to inspect residual plots during model evaluation to ensure linear regression assumptions are met.

A table with the calculation results and the plotted graphs are provided:

Variable	Mean	SD	Median	Q1 (25%)	Q3 (75%)	Min
progression	152.13	77.09	140.5	87	211.5	25
bmi (kg/m^2)	26.38	4.42	25.7	23.2	29.28	18
bp (mm Hg)	94.65	13.83	93	84	105	62
age (years)	48.52	13.11	50	38.25	59	19
s1 (tc)	189.14	34.61	186	164.25	209.75	97
s2 (ldl)	115.44	30.41	113	96.05	134.5	41.6
s3 (hdl)	49.79	12.93	48	40.25	57.75	22
s4 (tch)	4.07	1.29	4	3	5	2
s5 (ltg)	4.64	0.52	4.62	4.28	5	3.26
s6 (glu)	91.26	11.5	91	83.25	98	58

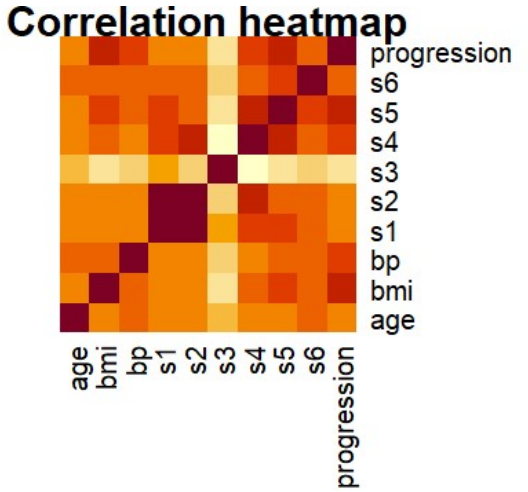




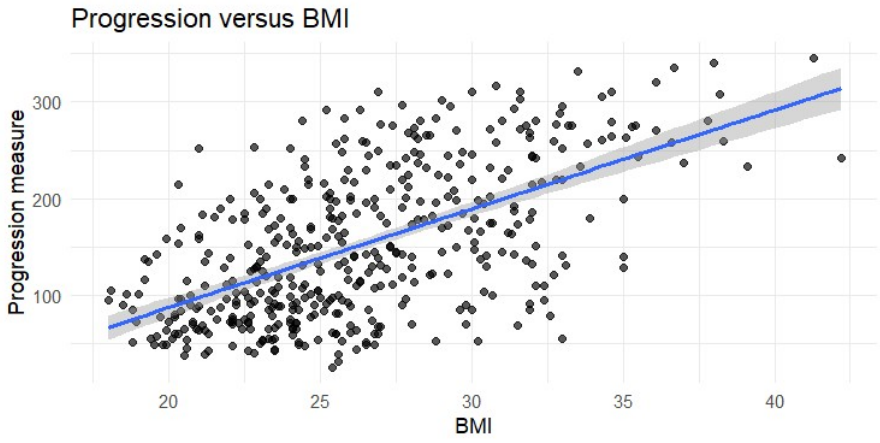
RELATIONSHIPS BETWEEN VARIABLES

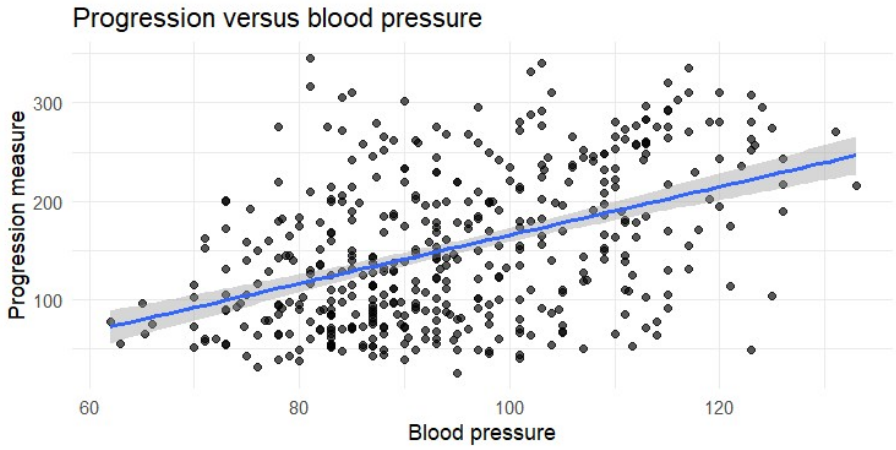
With the completion of the data's nature analysis, multicollinearity needs to be addressed, as to build a stable multiple regression model, including highly correlated variables together should be avoided. These important relations are monitored in the

Correlation heatmap, while the correlation between two variables is shown in scatterplots.



As shown in the heatmap, the two serum markers S1 and S2 carry heavily overlapping biological information (large dark square near the center of the map), while BMI, S5 and blood pressure are addressed as the strongest predictors. The relationship between two of these variables (BMI and blood pressure) and progression is depicted as genuinely linear in the scatterplots.





The steepness of the OLS line on the Progression versus BMI graph relative to the Pressure versus blood pressure graph reflects that BMI is the strongest predictor when comparing the two of them. This discrepancy is because, in the first graph, the data points cluster reasonably closely around the line instead of the wide visual dispersion on the second one, which indicates that blood pressure leaves a lot of variance unexplained. It is worth mentioning that while the upward slopes clearly show that higher BMI and higher blood pressure are associated with worse disease progression, this observational data does not prove that manually lowering a patient's BMI or BP will automatically halt the disease, as other unmeasured factors also affect the odds.

REGRESSION MODELING

In the simple Linear Regression model where only BMI is used, the fitted equation is

Progression = -117.77 + 10.23*BMI

The slope is 10.23. This means that for every 1-unit increase in a patient's BMI, their predicted disease progression score one year later increases by 10.23 points on average. The simple model has an R-squared of 0.3439, an adjusted R-squared of 0.3424, and a typical prediction error (sigma) of 62.52. This is explained by the sample data and it means that around 34% of the variance in disease progression among these patients can be explained by their BMI alone.

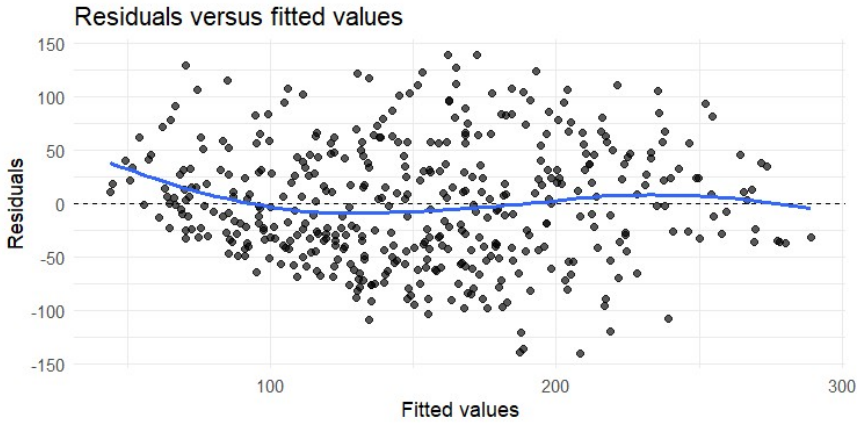
On the other side, Multiple Linear Regression which considers BMI S5 and blood pressure, improves the model's performance. The quality measurements show variations compared to the simple linear model: R-squared increases to 0.4801, adjusted R-squared increases to 0.4765 and typical prediction error drops to 55.78, while the effect of BMI decreases.

The most trustworthy metric to confirm the model's improvement is the adjusted R-squared. A larger raw R-squared is not automatically better because adding any predictor to a model mathematically forces R-squared to increase, even if the predictor is useless. The increased adjusted R-squared confirms that the added variables provided real explanatory power.

	R-SQUARED	ADJUSTED R-SQUARED	PREDICTION ERROR
SINGLE REGRESSION	0.3439	**0.3424**	62.52
MULTIPLE REGRESSION	0.4801	**0.4765**	55.78

MODEL EVALUATION

The residuals-versus-fitted plot provides important information on the model's analysis. The smoothed line deviates from the straight line ("U-shapped"), indicating potential non-linearity. Additionally, the fanning spread indicates heteroscedasticity (unequal error variance) which is expected, given the right-skewed distribution of the original disease progression variable. No multicollinearity is detected, as the VIF values are low (BMI= 1.34, BP= 1.27, S5= 1.34).



Using the correlation matrix in Question 3 excluded highly overlapping variables. As a result, the chosen predictors carry independent information. and the stability of the regression coefficients is trustworthy. The estimated coefficient of BMI is 6.50 with a 95% confidence interval of [5.13, 7.87]. The t-value is large (t = 9.32), yielding a p-value close to zero (<0.001). These results confirm with high statistical certainty that BMI is positively associated with progression. On the other side, the estimated coefficient of S5 is 49.58 with a 95% confidence interval of [38.00, 61.15]. The t-value is massive (t = 8.42, p < 0.001).

Statistical significance (the extremely low p-values) proves that an association is not possible to be zero. Practical importance, however, asks whether the size of the effect matters in the real world. If a 6.50-point change on the progression scale translates to barely noticeable symptoms for the patient, the finding- while statistically significant- might lack practical clinical importance.

CONCLUSION

Correlation analysis and multiple regression modeling identify Body Mass Index, log serum triglycerides (S5), and blood pressure (bp) as the most robust independent predictors of disease progression. While other serum markers (S1 and S2) strongly correlate with the outcome, their severe intercorrelation (r = 0.90) introduces multicollinearity, making them redundant for simultaneous modelling.

The final multiple regression model significantly outperforms a simple BMI-only model, explaining approximately 48% of the variance in disease progression (adjusted

R-squared = 0.477). Within this model, holding other factors constant, every 1-unit increase in BMI is associated with a 6.50-point increase in disease progression, demonstrating high statistical certainty.

Despite the model's high statistical significance, over half of the variance (52.3%) in disease progression remains unexplained. This points heavily to omitted variable bias; unmeasured critical lifestyle factors such as dietary habits, exercise routines, and concurrent medical treatments are likely influencing the outcome. Furthermore, residual diagnostics reveal mild heteroscedasticity, indicating that the model's predictive precision decreases for patients experiencing the most severe disease progression.

These findings represent mathematical correlations rather than proven mechanistic effects. It cannot be assumed that clinically intervening to lower a patient's BMI or blood pressure will automatically halt their disease progression at the rates estimated by the model coefficients.

Because this analysis utilizes sensitive data, GDPR principles need to be respected, prioritizing patient anonymity, secure storage, and clear purpose limitation.